Grade 9 Study Notes – Maths, Geography, Technology

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These notes follow the standard Grade 9 CAPS content for each subject. Check them against your school's exam guidelines.

Mathematics

Numbers, exponents and scientific notation

Number systems

Set	Symbol	Contains	Examples
Natural numbers	$\mathbb{N}$	Counting numbers	1; 2; 3; …
Whole numbers	$\mathbb{N}_0$	Natural numbers and 0	0; 1; 2; 3; …
Integers	$\mathbb{Z}$	Whole numbers and their negatives	…; −2; −1; 0; 1; 2; …
Rational numbers	$\mathbb{Q}$	Can be written as a fraction a/b (b ≠ 0); terminating or recurring decimals	½; −3; 0.75; 0.333…
Irrational numbers	$\mathbb{Q}'$	Cannot be written as a fraction; non-recurring, non-terminating decimals	√2; √5; π
Real numbers	$\mathbb{R}$	All rational and irrational numbers	

- **Prime number:** exactly two factors, 1 and itself (2, 3, 5, 7, 11…). 1 is not prime; 2 is the only even prime.
- **HCF** (highest common factor) and **LCM** (lowest common multiple): write each number as a product of prime factors. HCF = common primes (lowest powers); LCM = all primes (highest powers).
  - $24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$ → HCF = $2^2 \times 3$ = **12**; LCM = $2^3 \times 3^2$ = **72**.

Integers

- Adding a negative = subtracting: 5 + (−3) = 2. Subtracting a negative = adding: 5 − (−3) = 8.

- Multiplying/dividing: same signs → **positive**; different signs → **negative**. (−4) × (−3) = 12; (−12) ÷ 4 = −3.
- **Order of operations (BODMAS):** Brackets, Of/Orders (powers, roots), Division and Multiplication (left to right), Addition and Subtraction (left to right).
  - $3 + 4 \times 2^2 = 3 + 4 \times 4 = 3 + 16 = \mathbf{19}$

Exponents

In a^n, *a* is the **base** and *n* is the **exponent**: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

Law	Rule	Example
Multiply same base	$a^m \times a^n = a^{m+n}$	$x^3 \times x^4 = x^7$
Divide same base	$a^m \div a^n = a^{m-n}$	$y^8 \div y^3 = y^5$
Power of a power	$(a^m)^n = a^{mn}$	$(x^2)^3 = x^6$
Power of a product	$(ab)^n = a^n b^n$	$(2x)^3 = 8x^3$
Power of a quotient	$(a/b)^n = a^n/b^n$	$(x/3)^2 = x^2/9$
Zero exponent	$a^0 = 1\ (a \neq 0)$	$5^0 = 1$; $(3xy)^0 = 1$
Negative exponent	$a^{-n} = 1/a^n$	$2^{-3} = 1/8$

- Watch the signs: $(-2)^4 = 16$ but $-2^4 = -16$; $(-2)^3 = -8$.
- Simplify coefficients and variables separately: $3x^2y \times 4x^5y^3 = 12x^7y^4$; $(2a^3)^4 = 16a^{12}$; $12x^6 \div 4x^2 = 3x^4$.

Squares, cubes and roots

- $\sqrt{49} = 7$ because $7^2 = 49$; $\sqrt[3]{64} = 4$ because $4^3 = 64$; $\sqrt[3]{(-27)} = -3$.
- Learn squares to $15^2 = 225$ and cubes to $6^3 = 216$.
- $\sqrt{(a \times b)} = \sqrt{a} \times \sqrt{b}$, e.g. $\sqrt{(16 \times 9)} = 4 \times 3 = 12$. But $\sqrt{(9 + 16)} = \sqrt{25} = 5$, **not** 3 + 4.

Scientific notation

A number written as $\mathbf{a \times 10^n}$, where $\mathbf{1 \leq a < 10}$ and n is an integer.

- Large numbers → positive exponent: 45 000 000 = $\mathbf{4.5 \times 10^7}$ (decimal moved 7 places left).
- Small numbers → negative exponent: 0.00032 = $\mathbf{3.2 \times 10^{-4}}$ (decimal moved 4 places right).
- Multiply: $(3 \times 10^4) \times (2 \times 10^3) = 6 \times 10^7$.

- Divide: $(8 \times 10^9) \div (2 \times 10^3) = 4 \times 10^6$.
- Adjust if a is not between 1 and 10: $25 \times 10^5 = 2.5 \times 10^6$.

Algebraic expressions, factorising and equations

Terms and vocabulary

In **$3x^2 - 5x + 7$**:

- There are **3 terms** (separated by + and −), so it is a **trinomial**. (1 term = monomial, 2 terms = binomial.)
- **Coefficient** of x^2 is 3; coefficient of x is −5; **constant** is 7.
- **Degree** = highest exponent = 2.
- **Like terms** have the same variables with the same exponents: $3x^2$ and $-7x^2$ are like terms; $3x^2$ and 3x are not. Only like terms can be added or subtracted: $4a + 3b - a + 2b = 3a + 5b$.

Multiplying (expanding)

Type	Rule	Example
Monomial × binomial	Multiply each term inside	$3x(2x - 5) = 6x^2 - 15x$
Binomial × binomial	Each term × each term (FOIL)	$(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = x^2 - 2x - 15$
Square of a binomial	$(a + b)^2 = a^2 + 2ab + b^2$	$(x + 4)^2 = x^2 + 8x + 16$
	$(a - b)^2 = a^2 - 2ab + b^2$	$(2x - 3)^2 = 4x^2 - 12x + 9$
Difference of squares	$(a + b)(a - b) = a^2 - b^2$	$(x + 7)(x - 7) = x^2 - 49$
Binomial × trinomial	Each term × each term	$(x + 2)(x^2 - x + 3) = x^3 + x^2 + x + 6$

Common mistake: $(x + 4)^2$ is **not** $x^2 + 16$ — don't forget the middle term.

Dividing

Divide each term by the monomial: $(12x^3 - 8x^2 + 4x) \div 4x = 3x^2 - 2x + 1$.

Factorising (the reverse of expanding)

Always look for a **common factor first**.

1. **Common factor:** $6x^2 + 9x = 3x(2x + 3)$; $5a^2b - 10ab^2 = 5ab(a - 2b)$.
2. **Difference of two squares:** $a^2 - b^2 = (a + b)(a - b)$.
   - $x^2 - 25 = (x + 5)(x - 5)$
   - $9y^2 - 16 = (3y + 4)(3y - 4)$
   - $2x^2 - 50 = 2(x^2 - 25) = 2(x + 5)(x - 5)$ (common factor first)
3. **Trinomial $x^2 + bx + c$:** find two numbers that **multiply to c** and **add to b**.
   - $x^2 + 7x + 12 \rightarrow 3 \times 4 = 12, 3 + 4 = 7 \rightarrow (x + 3)(x + 4)$
   - $x^2 - x - 12 \rightarrow -4 \times 3 = -12, -4 + 3 = -1 \rightarrow (x - 4)(x + 3)$
   - $x^2 - 8x + 15 \rightarrow -3 \times -5 = 15, -3 + (-5) = -8 \rightarrow (x - 3)(x - 5)$

Check a factorisation by expanding it again.

Simplifying algebraic fractions

Factorise top and bottom, then cancel common **factors** (never cancel terms): $(x^2 - 9)/(x + 3) = (x + 3)(x - 3)/(x + 3) = x - 3$.

Solving equations

An equation is solved when you find the value of the variable that makes both sides equal. Whatever you do to one side, do to the other.

Type	Example	Solution
Linear	$3x - 7 = 11$	$3x = 18 \rightarrow$ **$x = 6$**
Variable on both sides	$5x + 2 = 2x - 10$	$3x = -12 \rightarrow$ **$x = -4$**
With brackets	$2(x - 3) = x + 4$	$2x - 6 = x + 4 \rightarrow$ **$x = 10$**
With fractions	$x/3 + x/2 = 5$	× LCD 6: $2x + 3x = 30 \rightarrow 5x = 30 \rightarrow$ **$x = 6$**
Product = 0	$(x - 2)(x + 5) = 0$	$x - 2 = 0$ or $x + 5 = 0 \rightarrow$ **$x = 2$ or $x = -5$**
Quadratic (factorise first)	$x^2 - 5x + 6 = 0$	$(x - 2)(x - 3) = 0 \rightarrow$ **$x = 2$ or 3**
Exponential	$2^x = 32$	$2^x = 2^5 \rightarrow$ **$x = 5$**

Check your answer by substituting it back into the original equation.

Word problems

1. Let x = the unknown.
2. Write an equation from the words.

3. Solve and answer in words.

Example: The sum of three consecutive numbers is 72. Let the numbers be x, x + 1, x + 2. Then 3x + 3 = 72 → x = 23. The numbers are **23, 24 and 25**.

Example: Thabo is 3 times as old as his sister. In 4 years the sum of their ages will be 32. Sister = x, Thabo = 3x → (x + 4) + (3x + 4) = 32 → 4x = 24 → x = 6. Sister is **6**, Thabo is **18**.

Patterns, functions and linear graphs

Number patterns

Linear pattern — a **constant difference** between terms. The general term is $\mathbf{T_n = dn + c}$, where d is the constant difference.

Example: 5; 8; 11; 14; …

- Constant difference d = 3, so $T_n = 3n + c$.
- Use $T_1 = 5$: 3(1) + c = 5 → c = 2. So $\mathbf{T_n = 3n + 2}$.
- $T_{50} = 3(50) + 2 = \mathbf{152}$.
- Which term equals 89? 3n + 2 = 89 → 3n = 87 → **n = 29** (the 29th term).

Shortcut: c = T_1 − d (the "zero term").

Geometric pattern — a **constant ratio** (multiply by the same number): 3; 6; 12; 24; … (× 2). $T_n = 3 \times 2^{n-1}$.

Patterns with pictures (e.g. matchsticks): count the items in each figure, find the constant difference, then the general rule. *Figure 1 = 4 sticks, figure 2 = 7, figure 3 = 10 → $T_n = 3n + 1$.*

Functions and relationships

A **function** gives exactly one output (y) for each input (x). It can be shown as a **flow diagram, table, formula (equation) or graph**.

x	−2	−1	0	1	2
y = 2x + 1	−3	−1	1	3	5

- **Input values** = x (independent variable). **Output values** = y (dependent variable).
- To find the rule from a table: the constant difference in y (when x goes up by 1) is the number in front of x; the y-value when x = 0 is the constant.

Linear graphs: $y = mx + c$

A linear function forms a **straight line**.

- **m = gradient (slope)** = steepness = rise ÷ run = $(y_2 - y_1) \div (x_2 - x_1)$.
  - m > 0: line goes **up** to the right. m < 0: line goes **down**. m = 0: horizontal line.
- **c = y-intercept** = where the line cuts the **y-axis** (put x = 0).
- **x-intercept** = where the line cuts the **x-axis** (put y = 0 and solve).

Example: $y = 2x - 4$

- Gradient = 2; y-intercept: x = 0 → y = −4 → point (0; −4).
- x-intercept: 0 = 2x − 4 → x = 2 → point (2; 0).
- **To draw it:** plot both intercepts and join them with a ruler, extending the line.

Example (gradient from two points): through (1; 3) and (4; 9): m = (9 − 3) ÷ (4 − 1) = 6 ÷ 3 = **2**.

Special lines

Equation	Line
y = k (e.g. y = 3)	**Horizontal** line through 3 on the y-axis; gradient 0
x = k (e.g. x = −2)	**Vertical** line through −2 on the x-axis; gradient undefined
y = mx (c = 0)	Passes through the **origin** (0; 0)

- **Parallel lines** have the **same gradient**: y = 3x + 1 and y = 3x − 5 are parallel.
- **Point of intersection** of two lines: set the equations equal and solve. y = x + 1 and y = −x + 5: x + 1 = −x + 5 → x = 2, y = 3 → **(2; 3)**.

Interpreting graphs

- **Increasing** (going up), **decreasing** (going down), **constant** (flat).
- **Linear** (straight) vs **non-linear** (curved).
- **Discrete** (separate points, e.g. number of people) vs **continuous** (joined line, e.g. time, temperature).
- On a distance–time graph, a steeper line means faster; a flat part means stopped.

Geometry, Pythagoras and transformations

In geometry questions, every statement needs a **reason** in brackets. Use the standard reasons below.

Angles and lines

Fact	Reason to write
Angles on a straight line add up to 180°	(∠s on a str line)
Angles around a point add up to 360°	(∠s round a pt)
Vertically opposite angles are equal	(vert opp ∠s)
Parallel lines: **corresponding** angles are equal (F-shape)	(corr ∠s; AB ∥ CD)
Parallel lines: **alternate** angles are equal (Z-shape)	(alt ∠s; AB ∥ CD)
Parallel lines: **co-interior** angles add up to 180° (U-shape)	(co-int ∠s; AB ∥ CD)

Types of angles: acute (< 90°), right (90°), obtuse (90°–180°), straight (180°), reflex (180°–360°), revolution (360°). **Complementary** angles add to 90°; **supplementary** angles add to 180°.

Triangles

- **Sum of angles** in a triangle = 180° (sum of ∠s of △).
- **Exterior angle** = sum of the two opposite interior angles (ext ∠ of △).
- **Isosceles** triangle: two equal sides, and the angles opposite them are equal (∠s opp = sides).
- **Equilateral** triangle: all sides equal, all angles 60°.
- **Scalene:** no equal sides. **Right-angled:** one 90° angle.

Congruent and similar triangles

Congruent (≡) = exactly the same shape and size. Conditions:

Condition	Meaning
SSS	Three sides equal
SAS	Two sides and the **included** angle equal

Condition	Meaning
AAS (or SAA)	Two angles and a corresponding side equal
RHS	Right angle, hypotenuse and one other side equal

(AAA and SSA do **not** prove congruency.)

Similar (|||) = same shape, different size. Conditions: **all corresponding angles equal** (AAA), or **all corresponding sides in the same ratio**.

Example: △ABC ||| △DEF with AB = 4, DE = 8, BC = 5. Ratio = 8 ÷ 4 = 2, so EF = 5 × 2 = **10**.

Write the letters in **corresponding order**: △ABC ≡ △DEF means A matches D, B matches E, C matches F.

Quadrilaterals

Shape	Properties
Parallelogram	Opposite sides parallel and equal; opposite angles equal; diagonals bisect each other
Rectangle	Parallelogram with four right angles; diagonals equal
Rhombus	Parallelogram with four equal sides; diagonals bisect at 90° and bisect the angles
Square	All sides equal, all angles 90°; diagonals equal and bisect at 90°
Trapezium	One pair of opposite sides parallel
Kite	Two pairs of adjacent sides equal; diagonals perpendicular; one diagonal bisects the other

Sum of the angles in a quadrilateral = **360°**. Sum of interior angles of any polygon = **180°(n − 2)**, n = number of sides (hexagon: 180 × 4 = 720°).

Theorem of Pythagoras

In a **right-angled triangle**: **hypotenuse² = sum of the squares of the other two sides**: $c^2 = a^2 + b^2$. The hypotenuse (c) is the longest side, opposite the right angle.

Find the hypotenuse: sides 6 cm and 8 cm. $c^2 = 6^2 + 8^2 = 36 + 64 = 100 \rightarrow c =$ **10 cm**.

Find a shorter side: hypotenuse 13, one side 5. $a^2 = 13^2 - 5^2 = 169 - 25 = 144 \rightarrow a =$ **12**.

Is it right-angled? Sides 7, 24, 25: $7^2 + 24^2 = 49 + 576 = 625 = 25^2$ → **yes**.

Useful triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25 (and multiples, e.g. 6-8-10).

Transformation geometry

Transformation	What happens	Rule on the Cartesian plane
Translation (slide)	Every point moves the same distance in the same direction	(x; y) → (x + a; y + b). 3 right, 2 down: (x; y) → (x + 3; y − 2)
Reflection in the x-axis	Mirror image across the x-axis	(x; y) → (x; −y)
Reflection in the y-axis	Mirror image across the y-axis	(x; y) → (−x; y)
Reflection in y = x	Swap x and y	(x; y) → (y; x)
Rotation 90° clockwise about the origin	Turn	(x; y) → (y; −x)
Rotation 90° anticlockwise about the origin	Turn	(x; y) → (−y; x)
Rotation 180° about the origin	Half turn	(x; y) → (−x; −y)
Enlargement / reduction by factor k from the origin	Shape gets bigger (k > 1) or smaller (0 < k < 1)	(x; y) → (kx; ky). Area is multiplied by **k^2**

Translations, reflections and rotations give **congruent** shapes. Enlargements give **similar** shapes.

Example: A(2; 5) reflected in the y-axis → A′(−2; 5). Rotated 180° → A′(−2; −5). Enlarged by factor 3 → A′(6; 15).

Perimeter, area, surface area and volume

Perimeter is the distance around a shape (units: cm, m). **Area** is the space inside a flat shape (cm^2, m^2). **Volume** is the space inside a 3D object (cm^3, m^3).

Area and perimeter of 2D shapes

Shape	Perimeter	Area
Square (side s)	4s	s^2
Rectangle (l × b)	2l + 2b	l × b
Triangle	sum of sides	½ × base × perpendicular height
Parallelogram	sum of sides	base × perpendicular height
Trapezium	sum of sides	½ × (sum of parallel sides) × height
Rhombus / kite	sum of sides	½ × diagonal_1 × diagonal_2
Circle (radius r)	**Circumference = 2πr** (or πd)	**πr²**

Use π ≈ 3.14 (or the calculator π key). Diameter = 2 × radius.

Example: circle with diameter 10 cm → r = 5. Circumference = 2 × 3.14 × 5 = **31.4 cm**; area = 3.14 × 5^2 = **78.5 cm²**.

Unit conversions

Length	Area	Volume / capacity
1 cm = 10 mm	1 cm^2 = 100 mm^2	1 cm^3 = 1 000 mm^3
1 m = 100 cm	1 m^2 = 10 000 cm^2	1 m^3 = 1 000 000 cm^3
1 km = 1 000 m		**1 cm^3 = 1 mℓ**; 1 000 cm^3 = 1 ℓ; 1 m^3 = 1 000 ℓ (1 kℓ)

Surface area of 3D objects

Surface area = the total area of all the faces (the **net** of the object).

Object	Surface area
Cube (side s)	$6s^2$
Rectangular prism (l, b, h)	2(lb + lh + bh)
Triangular prism	2 × area of triangle + (perimeter of triangle × length)
Cylinder (radius r, height h)	$2\pi r^2 + 2\pi rh$ (two circles + the curved rectangle)

Example (rectangular prism 5 cm × 3 cm × 4 cm): SA = 2(5×3 + 5×4 + 3×4) = 2(15 + 20 + 12) = 2 × 47 = **94 cm²**.

Example (cylinder r = 7 cm, h = 10 cm, π = 3.14): SA = 2(3.14)(7²) + 2(3.14)(7)(10) = 307.72 + 439.6 = **747.32 cm²**.

An **open** container (no lid) has one fewer face: open cylinder = $\pi r^2 + 2\pi rh$.

Volume of 3D objects

For any **prism**: **Volume = area of the base × height**.

Object	Volume
Cube	s^3
Rectangular prism	l × b × h
Triangular prism	(½ × base × height of triangle) × length of prism
Cylinder	$\pi r^2 \times h$

Example: rectangular box 5 cm × 3 cm × 4 cm: V = 5 × 3 × 4 = **60 cm³** = 60 mℓ.

Example: cylinder r = 7 cm, h = 10 cm: V = 3.14 × 49 × 10 = **1 538.6 cm³** ≈ 1.54 ℓ.

Example: triangular prism, triangle base 6 cm, triangle height 4 cm, prism length 10 cm: V = (½ × 6 × 4) × 10 = 12 × 10 = **120 cm³**.

Effect of changing dimensions

If all lengths are multiplied by k: **perimeter × k**, **area × k²**, **volume × k³**.

Example: doubling the sides of a cube makes its surface area 4 times bigger and its volume 8 times bigger.

Financial maths, data handling and probability

Financial maths

Term	Meaning
Principal (P)	Amount invested or borrowed at the start
Interest	Extra money earned (investment) or paid (loan)
Rate (i)	Interest rate per year, as a decimal (8% = 0.08)

Term	Meaning
n	Number of years
Accumulated amount (A)	Principal + interest at the end

Simple interest — interest is calculated on the **original amount only**: **$A = P(1 + i \times n)$**

Example: R5 000 at 8% p.a. simple interest for 3 years. $A = 5\,000(1 + 0.08 \times 3) = 5\,000 \times 1.24$ = **R6 200**. Interest = R1 200.

Compound interest — interest is calculated on the **principal plus interest already earned**: **$A = P(1 + i)^n$**

Example: R5 000 at 8% p.a. compound interest for 3 years. $A = 5\,000(1.08)^3$ = **R6 298.56**. Compound interest earns more than simple interest over time.

Hire purchase (HP) — buying on credit: pay a deposit, then monthly instalments with simple interest on the balance.

Example: A TV costs R12 000. Deposit 10%, balance paid over 2 years at 15% p.a. simple interest.

1. Deposit = 10% × 12 000 = R1 200 → balance = R10 800
2. Interest = 10 800 × 0.15 × 2 = R3 240 → amount owed = R14 040
3. Monthly instalment = 14 040 ÷ 24 = **R585**
4. Total paid = 1 200 + 14 040 = **R15 240** (R3 240 more than the cash price)

Other financial terms:

- **Profit** = selling price − cost price; **loss** if the selling price is lower. % profit = profit ÷ cost price × 100.
- **Discount** = reduction in price. 20% off R250 = R250 − R50 = R200.
- **VAT** in South Africa is **15%**. Price incl. VAT = price excl. VAT × 1.15. Price excl. VAT = price incl. VAT ÷ 1.15.
- **Inflation** = the general rise in prices over time (use the compound interest formula).
- **Exchange rates:** if $1 = R18, then $50 = 50 × 18 = R900, and R360 = 360 ÷ 18 = $20.
- **Budget:** a plan of income and expenses. **Income** − **expenses** = savings (surplus) or a shortfall (deficit).

Data handling

The data cycle: pose a question → collect data (questionnaire, survey, observation, experiment) → organise (tally table, frequency table, stem-and-leaf) → represent (graphs) → analyse → interpret and report.

- **Population** = the whole group; **sample** = a part of the population that is studied. A sample must be **representative** and not **biased**.
- **Categorical** data (colours, sports) vs **numerical** data: **discrete** (counted, e.g. number of siblings) or **continuous** (measured, e.g. height).

Measures of central tendency and spread — data: 4; 7; 7; 9; 13

Measure	How to find it	Answer
Mean (average)	Sum ÷ number of values	40 ÷ 5 = **8**
Median	Middle value of the **ordered** data (even number of values: average of the two middle ones)	**7**
Mode	Value that occurs most often	**7**
Range	Highest − lowest	13 − 4 = **9**

An **outlier** is a value far from the rest; it affects the mean most, so the median is better when there are outliers.

Graphs:

- **Bar graph** — categorical data; gaps between bars.
- **Histogram** — grouped numerical data; no gaps.
- **Pie chart** — parts of a whole. Angle of a sector = (frequency ÷ total) × 360°.
- **Line graph** — changes over time.
- **Scatter plot** — relationship between two variables (positive, negative or no correlation).
- Watch for **misleading graphs**: axes not starting at zero, uneven scales, missing labels.

Probability

Probability = how likely an event is, from **0 (impossible)** to **1 (certain)**.

P(event) = number of favourable outcomes ÷ total number of possible outcomes

- A die: P(even number) = 3 ÷ 6 = ½; P(7) = 0.
- P(event does **not** happen) = 1 − P(event).
- A **sample space** lists all possible outcomes.

Two events together — use a table or tree diagram.

Example: Toss a coin and roll a die. Sample space = 2 × 6 = **12** outcomes (H1, H2, …, T6). P(heads and a 6) = 1 ÷ 12.

Example: Toss two coins: HH, HT, TH, TT. P(one head and one tail) = 2 ÷ 4 = ½.

Theoretical vs relative frequency:

- **Theoretical probability** — what should happen (P(heads) = ½).
- **Relative frequency** — what actually happened in an experiment = number of times event occurred ÷ number of trials. 30 heads in 50 tosses → 30 ÷ 50 = 0.6.
- The more trials you do, the closer the relative frequency gets to the theoretical probability.

Geography

Map skills

Types of maps

Map	Description
Topographic map	Shows natural and man-made features with symbols and contour lines. South African topographic maps are **1:50 000**
Orthophoto map	An aerial photograph with contour lines and names added; scale **1:10 000** (shows more detail of a smaller area)
Atlas maps	Small-scale maps of large areas (countries, the world)
GIS (Geographic Information System)	A computer system that captures, stores, analyses and displays spatial (location) data in **layers** (e.g. roads, rivers, schools)

Scale

Scale shows the relationship between a distance on the map and the real distance on the ground.

Type	Example
Word scale	"1 cm represents 0.5 km"
Ratio (numerical) scale	1 : 50 000 (1 cm on the map = 50 000 cm on the ground = 500 m)

Type	Example
Line (linear) scale	A ruler-like bar divided into km

- **Large scale** (e.g. 1:10 000) = small area, lots of detail. **Small scale** (e.g. 1:1 000 000) = large area, little detail.
- **Calculating real distance:** measure on the map in cm, then multiply by the scale. On a 1:50 000 map, **1 cm = 0.5 km**, so 6 cm = 6 × 0.5 = **3 km**. On a 1:10 000 orthophoto map, **1 cm = 100 m (0.1 km)**.
- For curved distances (roads, rivers) use a piece of string or the edge of a paper.

Location: grid references (latitude and longitude)

- **Latitude** lines run east–west, measured **north or south of the Equator** (0° to 90°).
- **Longitude** lines run north–south, measured **east or west of the Prime (Greenwich) Meridian** (0° to 180°).
- South Africa is **south** of the Equator and **east** of Greenwich, so coordinates look like **33°45′S; 18°58′E** (Paarl area).
- 1 degree (°) = 60 minutes (′); 1 minute = 60 seconds (″). **Always give latitude first.**

Direction and bearing

- **16 compass directions:** N, NNE, NE, ENE, E, ESE, SE, SSE, S, SSW, SW, WSW, W, WNW, NW, NNW.
- Direction is always given **from** one place **to** another ("B is NE of A").
- **True bearing** = the angle measured **clockwise from true north** (0°–360°). East = 90°, South = 180°, West = 270°.
- **Magnetic bearing** = bearing measured from **magnetic north** (where a compass points). Magnetic declination is the angle between true north and magnetic north; it changes every year.

Contour lines and landforms

Contour lines are brown lines joining places of the **same height above sea level**. The **contour interval** on 1:50 000 maps is **20 m**.

Contour pattern	Landform / slope
Lines **close together**	**Steep** slope
Lines **far apart**	**Gentle** slope

Contour pattern	Landform / slope
Evenly spaced	**Even** (uniform) slope
Close at the top, far apart at the bottom	**Concave** slope
Far apart at the top, close at the bottom	**Convex** slope
Closed circles, highest in the middle	**Hill / conical hill**
Closed shape with a flat top, steep sides	**Mesa** (wide) / **butte** (narrow)
V-shapes pointing **up** to higher ground	**Valley** (river flows in it)
V-shapes pointing **down** to lower ground	**Spur**
Low area between two hills	**Saddle / nek**

- **Spot height** = a dot with the exact height. **Trigonometrical station** = a triangle with the height (survey point).
- **Gradient** = vertical interval (VI) ÷ horizontal equivalent (HE), written as 1 : x. VI = height difference; HE = real distance (same units!). *Example:* VI = 100 m, HE = 2 000 m → gradient = 100 ÷ 2 000 = **1 : 20**.
- **Cross-section:** a side view of the landscape along a line between two points, drawn from the contour heights.
- **Intervisibility:** two points can see each other if no higher ground lies between them (check on a cross-section).

Map symbols and conventional signs

Learn the key (legend): roads (national, main, secondary), railways, rivers (perennial = flows all year; non-perennial = dry part of the year), dams, buildings, schools, churches, power lines, cultivated land, orchards, vegetation, cemeteries, airfields.

Development issues

Development is the process of improving people's **quality of life** — not only income, but also health, education, housing, safety and freedom.

Measuring development (indicators)

Indicator	What it measures	Developed country	Developing country
GDP / GNI per capita	Average income per person per year	High	Low
Life expectancy	Average number of years a person is expected to live	High (80+)	Lower (50–65)
Literacy rate	% of adults who can read and write	Near 100%	Lower
Infant mortality rate	Babies who die before age 1, per 1 000 births	Low	High
Population growth rate	How fast the population grows	Slow	Fast
Access to clean water and sanitation	% of people with safe water and toilets	Almost all	Fewer
HDI (Human Development Index)	Combines income, education and life expectancy; a value from 0 to 1	Close to 1	Lower

GDP = total value of goods and services produced in a country in a year. **Per capita** = per person.

- **Economic indicators** measure wealth (GDP, income). **Social indicators** measure quality of life (health, education, housing).
- One indicator alone is not enough: an average hides **inequality** (South Africa has one of the highest levels of inequality in the world).

Developed and developing countries

- **Developed** ("economically more developed", the **North**): USA, UK, Germany, Japan, Australia.
- **Developing** ("less developed", the **South**): most of Africa, parts of Asia and South America.
- **Emerging economies** (developing fast): South Africa, Brazil, India, China (with Russia these form **BRICS**).

Factors that affect development

Factor	Effect
History (colonialism, apartheid)	Resources were taken; many people were denied education and land
Trade	Developing countries export cheap **raw materials** and import expensive **manufactured goods** (unfair trade)
Debt	Money spent on repaying loans cannot be spent on schools and hospitals
Corruption and poor governance	Money meant for development is stolen or wasted
Conflict and war	Destroys infrastructure and stops development
Natural disasters, climate, disease	Droughts, floods, HIV/AIDS and malaria slow development
Education and skills	An educated workforce attracts investment and earns more
Infrastructure	Roads, electricity, water and communication are needed for businesses
Population growth	Fast growth puts pressure on services and jobs

Trade

- **Exports** = goods sold to other countries; **imports** = goods bought from other countries.
- **Balance of trade** = exports − imports. A **surplus** if exports are more; a **deficit** if imports are more.
- **Fair trade** = producers in developing countries get a fair price and decent working conditions.
- **Trade barriers:** tariffs (taxes on imports), quotas (limits on amounts).
- **Adding value** (processing raw materials before exporting, e.g. turning cocoa into chocolate) earns more money and creates jobs.

Development aid

Type	Example	Advantage / disadvantage
Emergency (short-term) aid	Food, tents and medicine after a disaster	Saves lives but doesn't solve long-term problems
Long-term (development) aid	Building schools, clinics, wells; training	Helps people become self-sufficient
Bilateral aid	One government to another	May come with **conditions** ("tied aid")
Multilateral aid	Through organisations such as the UN, World Bank	Larger scale
NGO aid	Red Cross, Oxfam, Gift of the Givers	Works directly with communities

Aid can create dependency if it replaces local production. The best aid helps communities help themselves (e.g. teaching skills, small business loans).

Sustainable development

Sustainable development meets the needs of people today **without harming the ability of future generations** to meet their needs. It balances **economic growth, social needs and the environment**. Examples: renewable energy, recycling, community food gardens, eco-tourism, protecting water sources.

Resource use and sustainability

A **natural resource** is anything from nature that people use: water, soil, air, minerals, forests, fish, wildlife, sunlight.

Types of resources

Type	Meaning	Examples
Renewable	Replaced naturally if used wisely	Water, soil, forests, fish, sun, wind
Non-renewable	Finite; takes millions of years to form, will run out	Coal, oil, natural gas, gold, platinum, iron ore
Flow resources	Never run out regardless of use	Sunlight, wind, tides

Over-use of renewable resources can make them non-renewable in practice (e.g. over-fishing, deforestation, soil erosion).

Water

South Africa is a **water-scarce** country: low and unreliable rainfall (average about 450 mm per year, below the world average), high evaporation, frequent droughts.

- **Uses:** agriculture (irrigation uses the most), domestic, industry, mining, power stations.
- **Problems:** droughts (e.g. Cape Town's "Day Zero" 2018), pollution (sewage, mining acid water, fertilisers), leaking pipes, unequal access.
- **Water management:** dams and inter-basin transfer schemes (e.g. **Lesotho Highlands Water Project** supplies Gauteng), water restrictions, fixing leaks, rainwater tanks, recycling grey water, drip irrigation, protecting wetlands.

Soil

Soil takes hundreds of years to form but can be lost in one storm.

- **Soil erosion** = removal of topsoil by water or wind.
- **Causes:** overgrazing, deforestation, removing vegetation, ploughing up and down slopes, poor farming, veld fires.
- **Effects:** less fertile land, lower crop yields, food shortages, dongas (gullies), silting of dams and rivers.
- **Soil conservation:** contour ploughing (along the contours), terracing on slopes, planting trees and cover crops, crop rotation, rotational grazing, stopping overgrazing, building small walls (gabions) in dongas.

Forests and biodiversity

- **Deforestation** = clearing forests for farming, timber, firewood or building.

- **Effects:** loss of habitats and **biodiversity** (the variety of plant and animal life), soil erosion, less rainfall, more CO_2 (climate change).
- **Solutions:** replanting (reforestation), sustainable logging, protected areas (national parks, nature reserves), using alternatives to firewood.

Minerals and energy

- South Africa is rich in minerals: **gold, platinum, coal, diamonds, iron ore, manganese, chrome**. Mining creates jobs and export income.
- **Problems of mining:** non-renewable, scars the landscape, mine dumps, **acid mine drainage** polluting water, air pollution, dangerous working conditions.
- **Coal** provides most of SA's electricity but causes air pollution and climate change. Alternatives: solar, wind, hydro.

Pollution

Type	Causes	Solutions
Air	Factories, power stations, vehicles, burning	Filters, cleaner fuels, public transport, renewable energy
Water	Sewage, factory waste, fertilisers, litter, mining	Treat waste water, laws, clean-ups
Land	Litter, illegal dumping, landfills, mine dumps	Reduce, reuse, recycle; proper waste removal

Sustainable use of resources

Sustainable use = using resources in a way that meets present needs without destroying them for the future.

- **The 3 Rs:** Reduce, Reuse, Recycle.
- Use renewable energy; conserve water and electricity.
- Protect ecosystems (wetlands, forests, marine protected areas); fishing quotas.
- Laws and organisations: Department of Forestry, Fisheries and the Environment; SANParks; WWF.
- **Conservation** = protecting and managing resources carefully; **preservation** = keeping something untouched.

Surface forces that shape the Earth

The Earth's surface is shaped by **internal forces** (from inside: volcanoes, earthquakes, folding, faulting) and **external (surface) forces** that break down and move rock: **weathering, erosion and deposition**.

Weathering

Weathering = the breaking down of rock **in place** (it does not move).

Type	How it works	Examples	Climate
Physical (mechanical)	Rock breaks into smaller pieces without changing its chemical make-up	**Frost action** (water freezes in cracks, expands, splits rock); **exfoliation** (heating and cooling cause outer layers to peel off like an onion); plant roots growing in cracks	Dry areas with big temperature ranges; cold mountains
Chemical	Minerals in the rock change or dissolve	**Carbonation** (rainwater with CO_2 dissolves limestone → caves); **oxidation** (iron rusts, rock turns red); **hydrolysis**	Hot, wet areas
Biological	Living things break rock	Roots, burrowing animals, lichens producing acids	All climates

Erosion, transport and deposition

- **Erosion** = the wearing away **and removal** of rock and soil by agents: **water (rivers), wind, ice (glaciers), waves (sea) and gravity**.
- **Transport** = moving the eroded material.
- **Deposition** = dropping the material when the agent loses energy (slows down) — forming new landforms.
- **Mass movement** = downslope movement of material due to gravity: landslides, mudflows, rockfalls, soil creep.

Rivers and river landforms

Stage	Features	Landforms
Upper course (source, mountains)	Steep gradient, fast flow, **vertical erosion** (cuts down)	**V-shaped valleys**, waterfalls, rapids, interlocking spurs
Middle course	Gentler gradient, **lateral erosion** (sideways)	Wider valleys, **meanders** (bends)
Lower course (to the mouth)	Very gentle gradient, slow flow, **deposition**	**Flood plains**, **oxbow lakes** (cut-off meanders), **levees**, **deltas** and estuaries at the mouth

- A meander has fast water and erosion on the **outside** bend (river cliff) and slow water and deposition on the **inside** bend (slip-off slope / point bar).
- **Drainage basin** (catchment area) = the area drained by a river and its tributaries; **watershed** = the high ground separating two basins; **tributary** = a smaller river joining the main river; **confluence** = where two rivers meet.

Wind (aeolian) landforms

In deserts and dry areas: **sand dunes** (deposition), **mushroom rocks** / pedestal rocks (wind erosion at the base), and **deflation hollows** (wind removes loose material).

Coastal landforms

Erosion (destructive waves)	Deposition (constructive waves)
Cliffs and **wave-cut platforms**	**Beaches**
Caves → arches → stacks → stumps (a headland is eroded in this order)	**Spits** (sand extending into the sea)
Headlands and bays (hard rock forms headlands; soft rock erodes into bays)	**Sandbars** and lagoons

Human impact and hazards

- People speed up erosion through **deforestation, overgrazing, mining, building on slopes** and poor farming.
- Hazards: **floods** (e.g. KwaZulu-Natal 2022), **landslides**, coastal erosion damaging property.

- Reduce the impact: plant vegetation on slopes, build retaining walls and gabions, avoid building on flood plains, early-warning systems.

Technology

The design process and graphic communication

The design process (IDMEC)

Step	What you do
I – Investigate	Understand the problem: research, existing products, interviews, gather information
D – Design	Write a **design brief**, list **specifications** and **constraints**, draw several **ideas** (sketches) and choose the best
M – Make	Plan the steps, tools and materials (**flow chart / planning sheet**), then build the model or product safely
E – Evaluate	Test the product against the specifications; suggest improvements
C – Communicate	Present the solution with drawings, a portfolio and reports

- **Design brief** = a short statement of the problem and what must be designed. *Example: "I will design and make a model of a hydraulic lift that can raise a 200 g load 10 cm."*
- **Specifications** = requirements the product **must** meet (size, function, materials, cost, appearance, safety).
- **Constraints** = limitations (time, money, materials, tools available).
- **Evaluation** checks: does it work? Does it meet the specifications? Is it safe, attractive, affordable, environmentally friendly?

Graphic communication: drawing types

Drawing	Description	Used for
Freehand sketch	Quick drawing without instruments	Initial ideas

Drawing	Description	Used for
Oblique drawing	Front view drawn true shape; depth lines at **45°**, drawn at **half length** (cabinet oblique)	Simple 3D views
Isometric drawing	3D view; vertical lines stay vertical; horizontal edges at **30°**; all lengths true size	Showing an object in 3D
Single-point perspective	Lines meet at **one vanishing point** on the horizon	Realistic views (e.g. a road, a room)
Two-point perspective	Two vanishing points	Realistic views of buildings
Orthographic projection	2D views of an object: **front view, top view (plan), side view**; **first-angle projection** in SA	Working drawings for manufacturing
Exploded view	Parts drawn separated along their axes	Showing how parts fit together

Drawing conventions

Line type	Use
Thick (dark) continuous line	Visible outlines and edges
Thin continuous line	Dimension lines, projection lines, construction lines
Dashed (hidden) line	Hidden edges (not visible from that view)
Chain line (long dash, short dash)	Centre lines (circles, symmetry)

- **Dimensions** are in **millimetres (mm)**; units are not written on each dimension. Dimension lines have arrowheads touching the projection lines; numbers are written above the line.
- **Scale:** 1:1 = full size, 1:2 = half size (reduction), 2:1 = twice the size (enlargement).
- First-angle orthographic: the **top view is placed below** the front view, and the **left view on the right** of the front view.
- Use a **title block**: name, title, scale, date, drawing number.

Safety in the workshop

Wear safety goggles and closed shoes, tie back long hair, use tools for their purpose, cut away from your body, keep the work area clean, report injuries, never use broken tools or electrical equipment with damaged cords.

Mechanical systems and control

A **mechanism** changes an input motion or force into a useful output motion or force. A **system** = **input → process → output** (with feedback/control).

Types of motion: rotary (round and round — a wheel), **linear** (straight line — a train), **reciprocating** (back and forth in a line — a saw, piston), **oscillating** (swinging in an arc — a pendulum, windscreen wiper).

Levers

A lever is a rigid bar that turns around a **fulcrum** (pivot). It has an **effort** (input force) and a **load** (output force).

Class	Arrangement (middle item)	Examples
1st class	**Fulcrum** in the middle (E – F – L)	Seesaw, scissors, crowbar, pliers
2nd class	**Load** in the middle (F – L – E)	Wheelbarrow, bottle opener, nutcracker
3rd class	**Effort** in the middle (F – E – L)	Tweezers, fishing rod, braai tongs, your forearm

Memory aid **"FLE 123"**: the thing in the **middle** is F for class 1, L for class 2, E for class 3.

Mechanical advantage (MA) = load ÷ effort = effort arm ÷ load arm.

- MA > 1: the lever **makes work easier** (less effort needed). 2nd-class levers always have MA > 1.
- 3rd-class levers have MA < 1 — more effort, but greater movement and speed.
- **Moments:** load × load arm = effort × effort arm (when balanced).

Example: A wheelbarrow carries a 600 N load 0.4 m from the wheel (fulcrum). Effort is applied 1.2 m from the wheel. Effort = (600 × 0.4) ÷ 1.2 = **200 N**. MA = 600 ÷ 200 = **3**.

Gears

Gears are toothed wheels that transfer rotary motion. The gear turned by the input is the **driver**; the gear it turns is the **driven** gear.

- Two meshing gears turn in **opposite directions**. Add an **idler gear** between them to make the driver and driven gear turn in the **same** direction (the idler doesn't change the speed).
- **Gear ratio** = number of teeth on driven ÷ number of teeth on driver.
- **Small driver → big driven gear:** output turns **slower** but with **more force (torque)** — used when climbing hills on a bicycle.
- **Big driver → small driven gear:** output turns **faster** with less force.
- **Speed of driven gear** = speed of driver × (teeth on driver ÷ teeth on driven).

Example: A 20-tooth driver turns at 90 rpm and meshes with a 60-tooth driven gear. Gear ratio = 60 ÷ 20 = **3 : 1**. Speed of driven = 90 × (20 ÷ 60) = **30 rpm**, three times more torque.

Gear type	What it does	Example
Spur gears	Parallel shafts	Clocks, gearboxes
Bevel gears	Change direction of rotation by **90°**	Hand drill, egg beater
Worm and wheel	Large speed reduction; 90°; cannot be driven backwards	Guitar tuning pegs, gates
Rack and pinion	Rotary ↔ **linear** motion	Car steering, sliding gate

Pulleys and belts, sprockets and chains

- Pulleys joined by a belt turn in the **same direction** (crossed belt → opposite direction). Belts can slip (a safety feature).
- **Speed ratio** = diameter of driven pulley ÷ diameter of driver pulley. A small driver and a large driven pulley → slower, more force.
- **Sprockets and chain** (bicycle) do not slip.
- **Lifting pulleys:** a single fixed pulley changes the direction of force (MA = 1); a block and tackle with more pulleys gives MA > 1 (MA ≈ number of supporting ropes).

Other mechanisms

Mechanism	Motion change	Example
Cam and follower	Rotary → reciprocating	Car engine valves, toys
Crank and slider	Rotary ↔ reciprocating	Car engine piston, sewing machine
Linkages	Change direction or size of motion	Reverse-motion linkage, bell-crank linkage (90°), parallel-motion linkage (toolbox)
Ratchet and pawl	Allows rotation in one direction only	Winch, ratchet spanner
Screw	Rotary → linear; large force	Car jack, vice

Hydraulics and pneumatics

- **Hydraulic systems** use a **liquid** (oil or water); liquids **cannot be compressed**, so force is transferred efficiently. Examples: car brakes, hydraulic jack, excavator, dentist chair.
- **Pneumatic systems** use **air (gas)**; gas **can be compressed**, giving a softer action. Examples: air brakes on trucks, jackhammer, bus doors, tyre pump.
- **Pascal's law:** pressure applied to a liquid in a closed system is transmitted **equally in all directions**.
- **Pressure = force ÷ area** (P = F ÷ A), in pascal (Pa) or N/m².
- **Mechanical advantage** of a hydraulic system = area of output piston ÷ area of input piston (output force ÷ input force).

Example: Input piston area = 2 cm², output piston area = 20 cm², input force = 10 N.

Pressure = 10 ÷ 2 = 5 N/cm² (the same everywhere)
Output force = 5 × 20 = **100 N** (MA = 10)
The output piston moves only **1/10** of the distance the input piston moves — you gain force but lose distance.

Advantages of hydraulics: large force from a small effort, smooth precise control.
Disadvantages: leaks, oil can be messy and flammable.

Electrical systems and control

An electric circuit needs an **energy source** (input), **conductors**, a **control** (switch) and an **output device** (bulb, buzzer, motor). See also the Physical Sciences notes on current, voltage and resistance.

Components

Component	Function
Cell / battery	Energy source (input)
Switch (push, toggle, slide, reed)	Controls the circuit (on/off)
Resistor	Limits current to protect components
Variable resistor (potentiometer)	Adjusts current (volume, dimmer)
LDR (light-dependent resistor)	Resistance **decreases in bright light**; used as a light sensor (street lights)
Thermistor	Resistance changes with **temperature**; used as a heat sensor (fire alarm, kettle)
Diode	Lets current flow in **one direction only**
LED (light-emitting diode)	Gives light; needs a resistor in series; uses little energy
Transistor	Electronic **switch** or **amplifier**; a small base current switches a larger current
Capacitor	Stores charge for a short time (timing circuits)
Relay	An electromagnetic switch: a small current switches a bigger circuit
Buzzer / bulb / motor	Output devices (sound, light, movement)
Fuse / circuit breaker	Safety: breaks the circuit if current is too high

Resistor colour code

The first two bands are digits, the third band is the **multiplier** (number of zeros), the fourth is the **tolerance**.

Colour	Digit	Multiplier
Black	0	× 1
Brown	1	× 10
Red	2	× 100
Orange	3	× 1 000
Yellow	4	× 10 000
Green	5	× 100 000
Blue	6	× 1 000 000
Violet	7	
Grey	8	
White	9	

Tolerance: **gold = ±5%**, **silver = ±10%**.

Memory aid: **B**lack **B**ears **R**oam **O**ver **Y**ellow **G**rass **B**ut **V**iolets **G**row **W**ild (0–9).

Example: brown, black, red, gold → 1, 0, × 100 = **1 000 Ω (1 kΩ) ±5%**. *Example:* yellow, violet, orange, silver → 4, 7, × 1 000 = **47 000 Ω (47 kΩ) ±10%**.

Ohm's law and power

- **V = I × R** (voltage in V, current in A, resistance in Ω).
- **Power P = V × I** (watts, W). *Example:* a 230 V kettle drawing 8 A uses 230 × 8 = **1 840 W**.
- Series: $R_t = R_1 + R_2$, current the same, voltage shared. Parallel: voltage the same, current shared, total resistance smaller.

Example: An LED needs 2 V and 20 mA (0.02 A) from a 9 V battery. The resistor must drop 9 − 2 = 7 V. R = V ÷ I = 7 ÷ 0.02 = **350 Ω**.

Logic gates

Logic gates make decisions using two states: **1 = on (high)** and **0 = off (low)**. They can be built from switches.

Gate	Output is 1 when…	Switch model
AND	**Both** inputs are 1	Two switches in **series**

Gate	Output is 1 when…	Switch model
OR	**At least one** input is 1	Two switches in **parallel**
NOT (inverter)	The input is 0 (output is the opposite)	
NAND	NOT AND — output is 0 only when both inputs are 1	
NOR	NOT OR — output is 1 only when both inputs are 0	

Truth tables

A	B	AND	OR	NAND	NOR
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

NOT: input 0 → output 1; input 1 → output 0.

Examples: a machine that starts only when both safety buttons are pressed = **AND**; a doorbell that rings from the front **or** back door = **OR**; a light that comes on when it is **not** light outside = **NOT** (with an LDR).

Electrical safety and the environment

- **Fuses** melt and **circuit breakers** trip when the current is too high, preventing fires.
- The **earth wire** (green and yellow) carries current safely to the ground if a fault occurs. The **earth leakage** unit cuts power if current leaks (e.g. through a person).
- Plug wiring (SA): **brown = live**, **blue = neutral**, **green/yellow = earth**.
- Dangers: overloading plugs, damaged cords, water near electricity, illegal connections.
- Dispose of batteries properly (toxic chemicals); save electricity; use renewable energy.